

Member ID: _____

Time: _____

Rank: _____



DEVICE CONFIGURATION & TROUBLESHOOTING (305)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. Every time you reboot your computer the time is incorrect. Setting the correct time resolves the problem while the computer is running, but after a reboot it is incorrect again. How would you begin to troubleshoot the problem?
 - a. Replace the system memory
 - b. Update drivers
 - c. Replace the CMOS battery
 - d. Reinstall windows
2. What networking tool will show you the full path from one device to another?
 - a. tracert
 - b. ping
 - c. ipconfig
 - d. Windows network troubleshooting wizard
3. What service automatically assigns IP addresses to devices on a network?
 - a. DNS
 - b. ICMP
 - c. NAT
 - d. DHCP
4. What type of printer is most likely to be used for printing receipts?
 - a. Inkjet
 - b. Laser
 - c. Thermal
 - d. 3D
5. Which technology uses wear leveling to increase the lifespan of the hardware component?
 - a. Hard disk drive
 - b. Solid state drive
 - c. Network drive
 - d. USB flash drive
6. What Linux command can be used to view a list of files and directories?
 - a. dir
 - b. ls
 - c. cd
 - d. list disk

7. What Windows command line utility can be used to view and manage drives and partitions?
 - a. diskpart
 - b. list disk
 - c. df
 - d. chkdsk

8. You accidentally selected the wrong permissions for an important website and it will not prompt you to choose again. What can you do to clear your saved settings?
 - a. Close and reopen your internet browser
 - b. Disable tracking requests
 - c. Clear your browser history
 - d. Clear your saved cookies and cache

9. Which of the following *cannot* be used to manually set an IP address in Windows?
 - a. Right click networking icon in system tray > open network and internet settings
 - b. Control panel > network and sharing center
 - c. ipconfig ipv4 set address
 - d. netsh interface ipv4 set address

10. What type of port is often used to connect portable systems to a dock, and is also a common cell phone charging port?
 - a. USB
 - b. USB-c
 - c. USB-micro
 - d. USB-mini

11. If you wanted to increase your computer's file storage capacity, what would you purchase?
 - a. A memory card
 - b. A hard drive
 - c. System memory
 - d. A storage controller

12. After powering it off, what is the first thing you should do before opening a portable system for servicing?
 - a. Remove the battery
 - b. Connect an anti-static strap to bare metal
 - c. Hold the power button down for 30 seconds to drain power
 - d. Remove the plastic bezel

13. Which type of printer uses a drum during printing?
 - a. Inkjet
 - b. Laser
 - c. Thermal
 - d. Dot matrix

14. What is a common symptom of a failing printer drum?
 - a. Low toner errors
 - b. Printing blank pages
 - c. Streaks, lines, or smudges on the page
 - d. Faded colors

15. What file can you use to help troubleshoot a Windows blue screen error?
 - a. Dump file
 - b. Page file
 - c. Audit log
 - d. Setup log

16. Which Linux command can be used to rename files?
 - a. mv
 - b. cp
 - c. ren
 - d. ls

17. How can you check real time CPU usage if your computer is running slow?
 - a. Event viewer
 - b. Services
 - c. System Information
 - d. Task Manager

18. You are troubleshooting intermittent performance issues. How can you collect performance data over a period of time for analysis?
 - a. Event viewer
 - b. Task Manager
 - c. System Information
 - d. Performance Monitor

19. You are attempting to boot to OS installation media and keep getting a no boot device found error message. What would you do first?
- Update BIOS
 - Load the one time boot menu and select your device
 - Edit your BIOS boot settings
 - Change boot mode from BIOS to UEFI
20. Every time your computer boots, it always attempts to PXE boot before loading Windows. You want to stop this to make boot faster. What do you do?
- Update BIOS
 - Load the one time boot menu and select your device
 - Edit your BIOS boot settings
 - Change boot mode from BIOS to UEFI
21. What type of memory is the most commonly used in consumer PCs today?
- DDR2
 - DDR3
 - DDR4
 - DDR5
22. What memory form factor is often found in laptops?
- UDIMM
 - RDIMM
 - NVDIMM
 - SODIMM
23. What type of hard drive bypasses the traditional storage bus to connect directly to the PCI bus for massively increased IO speed?
- NVME
 - SSD
 - PCIE
 - HDD
24. What is the purpose of the default gateway?
- Separate network address from host address
 - Location to route traffic outside the local network
 - Device that provides DNS lookup services
 - Device that provides DHCP service

25. Which of the following is true about anti-static bags?
- a. The outside of the bag protects against ESD damage
 - b. Multiple components can be safely stored inside the same bag
 - c. Anti-static bags are no longer protective if they become wrinkled or worn
 - d. None of these are correct
26. What is the default file system for the majority of modern Linux distributions?
- a. Ext4
 - b. NTFS
 - c. XFS
 - d. FAT32
27. What type of drive is connected using ribbon cables?
- a. SATA
 - b. SAS
 - c. NVME
 - d. IDE
28. What should you be careful to *not* touch when replacing system memory?
- a. The memory chips
 - b. The golden contacts
 - c. The edges
 - d. All of these are correct
29. You should never open ____ due to risk of serious injury.
- a. an LCD monitor
 - b. a fan
 - c. a power supply
 - d. all of these are correct
30. Which utility scans your drive for file corruption and repairs it if found?
- a. sfc
 - b. chkdsk
 - c. defrag
 - d. diskpart
31. Which of the following common maintenance tasks would you *not* want to run against an SSD?
- a. defrag
 - b. chkdsk
 - c. sfc
 - d. Windows backup

32. What command can force your computer to get a new automatic IP address?
- a. ipconfig /release
 - b. ipconfig /renew
 - c. ipconfig /flushdns
 - d. netsh winsock reset
33. Which of the following is a class B subnet mask?
- a. 255.255.255.0
 - b. 255.0.0.0
 - c. 255.255.0.0
 - d. 255.255.255.255
34. What is the purpose of a riser card?
- a. Providing more network ports
 - b. Providing better graphics
 - c. Providing more storage
 - d. Providing more PCI slots
35. How much thermal grease should be used when replacing a CPU?
- a. A pea-sized amount
 - b. A dime-sized amount
 - c. A nickel-sized amount
 - d. Cover the entire top of the CPU
36. What type of backup will only backup files that have changed since the last full back up?
- a. Full
 - b. Incremental
 - c. Differential
 - d. Periodic
37. Your computer has a C drive for Windows files and a D drive for data. After a drive failure you lose all data on both the C and D drives. What happened?
- a. Both drives failed at the same time
 - b. Both drives were partitions on the same disk
 - c. The drives were part of a RAID
 - d. The files became corrupted

38. What does the OS watchdog timer error indicate?
- a. The system clock is losing time
 - b. The computer has a virus
 - c. The system crashed
 - d. The system files are corrupted
39. While attempting to install Windows you get an error message that no hard drives were found. What do you need to do to resolve the issue?
- a. Install a hard drive that is compatible with Windows
 - b. Install a different version of Windows
 - c. Recreate your installation media
 - d. Provide device drivers for your storage
40. How can you install Hyper-V on Windows 10?
- a. Download the install file from Microsoft and run it
 - b. Boot to your installation media and reinstall Windows with Hyper-V
 - c. Use Software Center
 - d. Use the Programs and Features utility
41. You want to format an external hard drive so it can be recognized by both Windows 10 and Ubuntu systems. What file system should you choose?
- a. NTFS
 - b. Ext4
 - c. XFS
 - d. None of these are correct
42. You can update device drivers through the ____ console.
- a. disk management
 - b. computer management
 - c. device management
 - d. driver management
43. Which of the following will *not* allow you to view and manage services in Windows?
- a. Computer Management
 - b. Disk Management
 - c. Task Manager
 - d. Administrative Tools

44. Smart card readers can be used for _____.
a. two factor authentication
b. processing debit card payments
c. biometric authentication
d. two factor authorization
45. What type of networking cable is most commonly used in home environments?
a. Fiber
b. Coax
c. CAT6
d. CAT5
46. Which command can be used to set the IP address in windows?
a. ipconfig
b. ping
c. netsh
d. ifconfig
47. What command can be used to view the IP address in Linux?
a. ipconfig
b. ping
c. netsh
d. ifconfig
48. Where should you obtain the latest device drivers for your system?
a. Manufacturer's website
b. Windows update
c. Disk that came with the computer
d. Google
49. Your computer is not powering on. What do you check first?
a. Power supply switch
b. Bad power supply
c. Broken power button
d. Power supply cable
50. Your computer is having trouble starting. You can see the power button light up and hear the fans spin. There are no error lights anywhere on the computer, but your screen is completely blank. What do you think the problem is?
a. The computer can't power on
b. The computer can't POST
c. The computer can't boot
d. The computer has no video